

Parthajit Das

Kolkata, India

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Education

Techno Main Salt Lake, West Bengal, India

Sept 2022 – June 2026

B.Tech in Computer Science and Engineering (CGPA: 8.50)

Relevant Coursework: Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Theory of Computation, Machine Learning

Central Modern School, Baranagar, West Bengal

March 2022

ISC Class XII: 90%

Central Modern School, Baranagar, West Bengal

March 2020

ICSE Class X: 94.2%

Experience

Research Trainee

June 2025 – July 2025

Tenth Summer School on Computer Vision, Image Processing, Machine Learning Vision Language Models *Electronics and Communication Sciences Unit, Indian Statistical Institute (ISI), Kolkata*

- Selected for an intensive program covering machine learning, deep learning, and computer vision.
- Completed a mentored project on **Generative Adversarial Networks**.
- Awarded **Special Mention (Top 3)** among participating teams.

Research & Project Experience

Hybrid Movie Recommendation System using Transformers and Dynamic Fusion

B.Tech Final Year Project

- Designed a hybrid recommendation framework integrating **DistilBERT-based content embeddings**, **Neural Collaborative Filtering (NCF)**, **SVD**, and **LightGBM ranking**, supported by a dynamic fusion strategy to combine semantic and collaborative signals for personalized recommendations.
- Modeled complex user-item interactions using embedding layers and **MLP-based neural architectures**.
- Addressed key recommender system challenges including **cold-start**, **data sparsity**, and **filter bubble effects**.
- Utilized **MovieLens** and **TMDB datasets** for user ratings, movie metadata, and semantic feature extraction.
- Evaluated system performance using **RMSE**, **Precision**, **Recall**, and **F1-Score**, achieving superior results compared to individual recommendation models.

Comparative Analysis of GAN Architectures

Tenth Summer School on Computer Vision, Image Processing, Machine Learning Vision Language Models

- Implemented and compared **Vanilla GAN**, **WGAN**.
- Trained models on MNIST, CIFAR-10, and Anime datasets.
- Evaluated using **FID**, **Inception Score**, and **qualitative analysis**.
- Observed improved stability and performance in WGAN-GP over Vanilla GAN.

Technical Skills

- Programming:** C/C++, Python, Java, SQL
- ML/DL:** PyTorch, TensorFlow, Transformers, LightGBM
- Libraries:** NumPy, Pandas
- Tools:** Git, GitHub, Kaggle, Google Colab, MySQL

Achievements

- Special Mention Award (Top 3)** – ISI Kolkata Summer School
Recognized among the top projects for “*Comparative Analysis of Generative Adversarial Network Architectures*”.
- AIR 2855 (98.7 percentile) in GATE CSE 2026 among 2,10,000+ candidates.
- 91.4 percentile in GATE CSE 2026 among 1,70,000+ candidates.

Publications / Manuscripts

- An Ensemble Learning Framework for Hybrid Movie Recommendation Systems Using Transformers**
Developed a research-oriented hybrid recommender system combining DistilBERT, Neural Collaborative Filtering (NCF), SVD, and LightGBM to improve recommendation accuracy and mitigate cold-start and data sparsity issues. Conducted comparative evaluation using RMSE, Precision, Recall, and F1-Score, demonstrating superior performance over individual recommendation models. Status: Unpublished Manuscript. Link: [View Manuscript](#)
- Comparative Analysis of Generative Adversarial Network Architectures**
Manuscript prepared during research training at ISI Kolkata Tenth Summer School, involving implementation and evaluation of

GAN variants (Vanilla GAN, WGAN, WGAN-GP).
Status: Unpublished (academic exposure).
Link: [View Manuscript](#)

Research Interests & Exposure

- **Research Interests:** Machine Learning, Deep Learning, Recommender Systems, Natural Language Processing, Generative Models.
- **Exposure:** Studied research literature in recommender systems, generative models, and transformer-based architectures, focusing on model design and evaluation metrics.